function: you don’t want to have the same identification number as your friend!

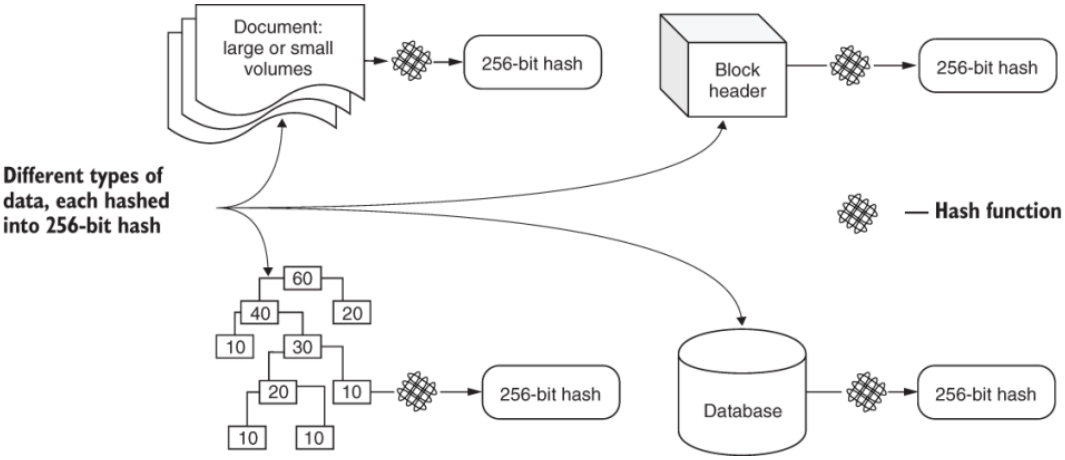


Figure 5.9 Transforming different types of data to 256-bit hashes

5.3.1 Digital signing of documents

For the digital signing of a document, a hash function is used to compute a hash of the document. This hash is used as the digital signature for the document and attached to the document by the sender. This signature can be verified later by the receiver of the document by recomputing the hash of the document and comparing it with the attached digital signature (hash).

5.3.2 Hashed data on distributed ledger

Blockchain is not your regular database; it stores only the minimal data needed in its distributed ledger. Hashing helps here too! The blockchain isn’t overloaded by a large document because only the hash value (representation) of the document can be stored on the chain. You’ll learn more about the versatility of hashing when we develop our decentralized system model further with on-chain and off-chain data in chapter 6.